OAK INTERNATIONAL SCHOOL

PREMIUM MATHEMATICS ACTIVITY WORKBOOK — CLASS 1

Grade Level: CM2 / IB MYP 1 (Ages 10–11)

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Programmes: IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

Welcome to Your Dual-Curriculum Mathematics Journey!

Bienvenue dans votre parcours mathématique d'excellence à Oak International School !

At Oak International School, you are developing mathematical superpowers by studying two world-class curricula at the same time:

1. The **International Baccalaureate (IB) Middle Years Programme (MYP 1)**, where you learn to inquire, explore patterns, communicate ideas in English, and see how mathematics shapes our global world.
2. The **Beninese National Curriculum (Approche Par Compétences — APC)**, where you develop lightning-fast mental arithmetic, rigorous step-by-step problem-solving in French, and deep understanding of practical calculations needed for the **Certificat d'Études Primaires (CEP)**.

How This Workbook Works

This workbook is your bridge between English and French mathematics. You will discover that whether you say "one half" or *"un demi"*, the mathematics is the same powerful language!

- Follow the instructions carefully for each section.
- Work neatly. In mathematics, clear presentation is part of clear thinking.
- Use the bilingual glossaries to learn technical words in both languages.
- Complete the working space provided. Always write your intermediate steps and units of measurement.

SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

Technical Lexicon: Numbers, Fractions, Operations & Mensuration

English Term	French Equivalent (*Terme Français*)	Mathematical Definition & Symbolic Example	Common False Friends & Translation Pitfalls
Whole number	*Nombre entier naturel*	A positive integer without decimals: 0, 1, 2, 3, \dots \in \mathbb{N}.	Do not confuse with "integer" which includes negative numbers (\mathbb{Z}).
Decimal mark / point	*Virgule décimale*	Symbol separating whole units from tenths: e.g., 12.5 (UK/IB) vs 12,5 (Benin).	In English use a dot (3.14); in French use a comma (3,14).
Numerator / Denominator	*Numérateur / Dénominateur*	In fraction \frac{a}{b}, a is the numerator (parts taken), b is denominator (total parts).	Pronunciation: *numérateur* is on top; *dénominateur* names the fractional unit.
Equivalent fractions	*Fractions égales / équivalentes*	Fractions with equal value: \frac{2}{3} = \frac{4}{6} = \frac{20}{30}.	Always state the multiplying factor applied to both terms.
Lowest terms / Irreducible	*Fraction irréductible*	A fraction where numerator and denominator share no common divisor > 1: e.g., \frac{3}{5}.	In French, say *« rendre irréductible »*, not "simplifier au plus bas".
Cost price / Selling price	*Prix de revient / Prix de vente*	*Prix de revient* = Purchase price + transport/overhead fees; *Bénéfice* = Selling price - Cost price.	*Frais* translates to "overheads" or "expenses", not "costs".
Area / Perimeter	*Aire (ou Surface) / Périmètre*	Perimeter is total boundary length (P in m); Area is total surface enclosed (A in \text{m}^2).	In English, "surface area" usually refers to 3D solids; for 2D, use "area".
Capacity / Volume	*Capacité / Volume*	Volume is the spatial extent (\text{m}^3, \text{dm}^3); Capacity is the fluid volume held (\text{L}, \text{cL}).	Crucial equivalence: 1\text{ dm}^3 = 1\text{ L} = 1\text{ kg} of pure water.
Hectare (ha)	*Hectare (ha)*	Agrarian unit of area equal to a square of 100\text{ m} \times 100\text{ m} = 10\ 000\text{ m}^2 = 1\text{ hm}^2.	Indispensable for Beninese agricultural CEP problems.

SECTION C — SHARED CONCEPTUAL FOUNDATIONS

Unit Theme: Fractions, Proportional Sharing and Commercial Mensuration

1. Conceptual Introduction: The Multiplicative Nature of Fractions

A fraction \frac{a}{b} represents two complementary mathematical operations:

1. **Partitioning**: Dividing a whole unit into b identical equal sub-parts and selecting a of those parts.
2. **Division Operator**: The exact quotient of a divided by b.

\frac{a}{b} = a \div b \quad (b \neq 0)

In both the IB MYP and the Beninese curriculum, a fraction is an operator that can scale a quantity:

\text{Fraction of a Quantity} = \frac{a}{b} \times Q = \frac{a \times Q}{b} = a \times \frac{Q}{b}

2. Visual Representation: The Number Line and Area Bar Model

Consider the fraction \frac{3}{4}:

- On a number line from 0 to 1, subdivide the unit interval into 4 segments of length 0.25. The third tick mark represents \frac{3}{4} = 0.75.
- In a rectangular area of 100\text{ m}^2, \frac{3}{4} represents three 25\text{ m}^2 quadrants = 75\text{ m}^2.

3. Worked Example: Method Comparison Across Programmes

Problem: An agricultural cooperative in Parakou harvests $1\ 200\text{ kg}$ of organic cashews. They sell $\frac{3}{5}$ of the harvest at 800 CFA F per kilogram. What is the total revenue?

- **The IB Method (Step-by-step Unitary Scaling)**:
- Find $\frac{1}{5}$ of the harvest: $1\ 200 \div 5 = 240\text{ kg}$.
- Multiply by the numerator: $240 \times 3 = 720\text{ kg}$ sold.
- Calculate revenue: $720\text{ kg} \times 800\text{ CFA F/kg} = 576\ 000\text{ CFA F}$.
- **The Beninese APC Method (Standard Formulated Solution)**:
- *Masse de cajou vendue* : $\frac{1\ 200 \times 3}{5} = \frac{3\ 600}{5} = 720\text{ kg}$.
- *Recette totale* : $720 \times 800 = 576\ 000\text{ CFA F}$.
- **Synthesis**: Both methods yield the identical outcome. The IB approach emphasizes unit rate thinking, while the Beninese APC focuses on the formal fractional operator and compact calculation.

SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

Activity 1: The Dantokpa Market Grain Merchant

1. Problem Situation

Madame Akpéné is a prominent grain merchant at the Dantokpa Market in Cotonou. She receives a delivery of 2.4 tonnes ($2\ 400\text{ kg}$) of high-grade maize from a farmer in Glazoué.

- Packaging: She packages $\frac{5}{8}$ of the maize into large 50 kg sacks for wholesale restaurants.
- Retail: She packages the remaining maize into small 2.5 kg retail bags for local family sales.
- Wholesale sacks sell for $18\ 000\text{ CFA F}$ each.
- Retail bags sell for $1\ 100\text{ CFA F}$ each.
- Transport overheads cost $45\ 000\text{ CFA F}$ for the entire shipment.

2. The IB Perspective (Inquiry & Representation)

- Identify the fraction of maize remaining for retail sales:

$$1 - \frac{5}{8} = \frac{3}{8}$$

- Construct a visual bar model partitioned into 8 equal units of 300 kg each.
- Wholesale portion (5 units): $5 \times 300\text{ kg} = 1\ 500\text{ kg}$.
- Retail portion (3 units): $3 \times 300\text{ kg} = 900\text{ kg}$.

3. The Beninese APC Perspective (Structured Solution / *Démarche Raisonnée*)

- *Quantité de maïs pour la vente en gros* :

$$\frac{2\ 400 \times 5}{8} = 1\ 500\text{ kg}$$

- *Nombre de sacs de 50 kg* :

$$1\ 500 \div 50 = 30\text{ sacs}$$

- *Recette de la vente en gros* :

$$30 \times 18\ 000 = 540\ 000\text{ CFA F}$$

- *Quantité de maïs pour la vente au détail* :

$$2\ 400 - 1\ 500 = 900\text{ kg}$$

- *Nombre de sachets de $2,5\text{ kg}$* :

$$900 \div 2,5 = 360\text{ sachets}$$

- *Recette de la vente au détail* :

$$360 \times 1\ 100 = 396\ 000\text{ CFA F}$$

- *Recette totale brute* :

$$540\ 000 + 396\ 000 = 936\ 000\text{ CFA F}$$

- *Bénéfice net après déduction des frais de transport* :

$$\text{Bénéfice} = 936\ 000 - 45\ 000 = 891\ 000\text{ CFA F}$$

4. Comparison of Methods

- IB uses fraction subtraction to verify partition consistency ($\frac{5}{8} + \frac{3}{8} = 1$).
- Benin APC uses direct sequential arithmetic reasoning with explicit labels (*phrase explicative*) for each intermediate result.

5. Student Reflection Question

Explain in two full sentences how calculating the unitary value of $\frac{1}{8}$ of the stock helps you verify both the wholesale mass and the retail mass.

Espace de réponse:

_____

_____

_____

_____

6. Application in a New Context

A pineapple cooperative in Allada harvests $3\ 600\text{ pineapples}$. They export $\frac{4}{9}$ to Europe by air cargo and process the remainder into bottled juice. How many pineapples are processed into juice?

Working space:

_____

_____

_____

_____

SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

Level 1: Foundational Understanding (Single-step operations)

1. Calculate without a calculator:

a) $45\ 670 + 128\ 455 =$

b) $300\ 000 - 142\ 850 =$

c) $1\ 250 \times 48 =$

d) $3\ 726 \div 6 =$

2. Convert into the requested metric unit:

a) $4.5\text{ km} = _____\text{ m}$

b) $3\ 500\text{ kg} = _____\text{ tonnes}$

c) $12.5\text{ L} = _____\text{ dm}^3 = _____\text{ cL}$

d) $2.5\text{ ha} = _____\text{ m}^2$

Level 2: Standard Application (Fractions & Perimeter/Area)

3. Simplify each fraction to its lowest irreducible form:

a) $\frac{24}{36} =$

b) $\frac{45}{60} =$

c) $\frac{120}{300} =$

4. Calculate and simplify:

a) $\frac{3}{7} + \frac{2}{7} =$

b) $\frac{7}{10} - \frac{3}{10} =$

c) $\frac{1}{4} + \frac{3}{8} =$

5. A rectangular garden plot in Calavi measures 28 m in length and 15 m in width.

a) Calculate its perimeter P.

b) Calculate its surface area A.

Level 3: Multi-Step Problems (Commercial Arithmetic & Proportionality)

6. A shopkeeper in Porto-Novo buys 15 crates of cooking oil for 180\ 000\text{ CFA F}. Transport costs are 7\ 500\text{ CFA F} for the delivery.

a) Calculate the total cost price (*prix de revient*) of the 15 crates.

b) If he sells each crate for 14\ 500\text{ CFA F}, calculate his total selling price.

c) Determine his net profit (*bénéfice*).

Level 4: Reasoning, Investigation & Unfamiliar Contexts

7. The Geometric Mosaic Pattern Investigation:

A tile designer in Ouidah creates square mosaic borders using black and white tiles:

- For a square border of side length 3 tiles: 8 border tiles are used.
- For a square border of side length 4 tiles: 12 border tiles are used.
- For a square border of side length 5 tiles: 16 border tiles are used.

a) Record the values in a table for side lengths s = 3, 4, 5, 6.

b) State the pattern rule in words.

c) Write a general algebraic rule for the number of border tiles B in terms of side length s.

d) Verify your rule for a square of side length 10 tiles.

Level 5: Extension & CEP Examination Challenge

8. Situation d'Évaluation Type CEP:

Monsieur Dossou possède un terrain rectangulaire de 120\text{ m} de long sur 80\text{ m} de large.

Il réserve \frac{2}{5} du terrain pour la culture du manioc et \frac{1}{4} pour la culture du maïs.

Le reste du terrain est consacré à la construction d'un poulailler et d'un réservoir d'eau.

1. Calcule la superficie totale du terrain en \text{m}^2 puis en hectares.
2. Calcule la superficie occupée par le manioc.
3. Calcule la superficie occupée par le maïs.
4. Calcule la superficie restante réservée au poulailler et au réservoir.
5. Pour clôturer entièrement son terrain, il pose 3 rangées de fil de fer en laissant une ouverture de 4\text{ m} pour le portail. Quelle longueur totale de fil de fer doit-il acheter ?

SECTION F — INTEGRATED APPLICATION TASKS

Task F-1: The Solar Irrigation Water Tower (MYP Criterion D & Benin CD 1/CD 2)

Context: A community agricultural project in Djougou installs an elevated cylindrical water storage tank fed by a solar pump to irrigate vegetable plots during the dry season.

- Tank dimensions: Circular base radius r = 1.4\text{ m}, Height h = 3\text{ m}.
- The daily water requirement for the crops is 4\ 500\text{ litres}.
- The solar pump operates from 08:00 to 16:00 (8 hours) and pumps at a constant rate of 25\text{ litres per minute}.

Student Questions:

1. Calculate the base area of the cylindrical water tank using \pi \approx \frac{22}{7}.
2. Calculate the total volume of the tank in cubic metres (\text{m}^3).
3. Convert this volume into litres, using the equivalence 1\text{ m}^3 = 1\ 000\text{ dm}^3 = 1\ 000\text{ litres}.
4. Calculate the total volume of water pumped by the solar system during an 8-hour sunny day.
5. Will the pump supply sufficient water for the daily crop requirements? Justify with exact numbers.

SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

Part G-1: IB MYP 1 Criterion B Assessment Task

Pattern Exploration: Growing Stepped Towers

Examine the pattern of towers made from centimeter cubes:

- Stage 1: 1 column of height 1 (1 cube).
- Stage 2: 2 columns of heights 1, 2 (3 cubes).
- Stage 3: 3 columns of heights 1, 2, 3 (6 cubes).
- Stage 4: 4 columns of heights 1, 2, 3, 4 (10 cubes).

1. Draw Stage 5 and determine the total number of cubes required.
2. Complete the table of values for Stages 1 to 6.
3. Describe the rule for finding the number of cubes needed for any stage n.
4. Calculate how many cubes are needed for Stage 12.

Part G-2: Beninese National Exam (CEP) Arithmetic Test

Épreuve d'Application Pratique

Un éleveur de Bohicon vend 8 bœufs à 275\ 000\text{ CFA F} l'unité.

Sur le montant perçu, il verse la moitié sur son compte d'épargne à la CLCAM.

Avec les \frac{3}{5} du reste, il achète des aliments concentrés pour son cheptel.

Avec la somme restante, il achète du grillage à 1\ 760\text{ CFA F} le mètre pour agrandir son enclos.

1. Calcule le montant total tiré de la vente des bœufs.
2. Calcule la somme déposée sur le compte d'épargne.
3. Calcule la somme dépensée pour les aliments concentrés.
4. Calcule la somme restante disponible.
5. Calcule la longueur de grillage que l'éleveur peut acheter.

SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

Diagnostic Pre-Flight Check (Self-Assessment)

Before starting this workbook, check your confidence (1 = Need help, 3 = Confident):

- [ ] I can multiply a 3-digit number by a 2-digit number without a calculator.
- [ ] I know that 1\text{ hectare} = 10\ 000\text{ m}^2.
- [ ] I can find equivalent fractions by multiplying or dividing top and bottom.
- [ ] I can calculate the perimeter and area of a rectangle.

COMPLETE WORKED SOLUTIONS & MARKING SCHEME

Solutions for Section E (Progressive Practice)

1. Arithmetic Calculations (1 mark each):

a) 45\ 670 + 128\ 455 = 174\ 125

b) 300\ 000 - 142\ 850 = 157\ 150

c) 1\ 250 \times 48 = 60\ 000

d) 3\ 726 \div 6 = 621

2. Metric Conversions (1 mark each):

a) 4.5\text{ km} = 4\ 500\text{ m}

b) 3\ 500\text{ kg} = 3.5\text{ tonnes}

c) 12.5\text{ L} = 12.5\text{ dm}^3 = 1\ 250\text{ cL}

d) 2.5\text{ ha} = 2.5 \times 10\ 000 = 25\ 000\text{ m}^2

3. Fractions Simplification (1 mark each):

a) \frac{24 \div 12}{36 \div 12} = \frac{2}{3}

b) \frac{45 \div 15}{60 \div 15} = \frac{3}{4}

c) \frac{120 \div 60}{300 \div 60} = \frac{2}{5}

4. Fraction Operations (1 mark each):

a) \frac{3+2}{7} = \frac{5}{7}

b) \frac{7-3}{10} = \frac{4}{10} = \frac{2}{5}

c) \frac{2}{8} + \frac{3}{8} = \frac{5}{8}

5. Calavi Garden (2 marks):

a) P = 2 \times (28 + 15) = 2 \times 43 = 86\text{ m}.

b) A = 28 \times 15 = 420\text{ m}^2.

6. Porto-Novo Merchant (3 marks):

a) *Prix de revient* = 180\ 000 + 7\ 500 = 187\ 500\text{ CFA F}.

b) *Prix de vente total* = 15 \times 14\ 500 = 217\ 500\text{ CFA F}.

c) *Bénéfice* = 217\ 500 - 187\ 500 = 30\ 000\text{ CFA F}.

7. Mosaic Investigation (4 marks):

a) Values: s=3 \implies 8; s=4 \implies 12; s=5 \implies 16; s=6 \implies 20.

b) Rule: Each step increases the side length by 1 and adds 4 border tiles.

c) Formula: B = 4(s - 1) or B = 4s - 4.

d) For s=10: B = 4(10 - 1) = 36 tiles.

8. Dossou Land CEP Task (5 marks):

1. *Superficie totale* : 120 \times 80 = 9\ 600\text{ m}^2 = 0,96\text{ ha}.
2. *Manioc* : \frac{9\ 600 \times 2}{5} = 3\ 840\text{ m}^2.
3. *Maïs* : \frac{9\ 600 \times 1}{4} = 2\ 400\text{ m}^2.
4. *Superficie restante* : 9\ 600 - (3\ 840 + 2\ 400) = 9\ 600 - 6\ 240 = 3\ 360\text{ m}^2.
5. *Clôture* : Périmètre = 2 \times (120 + 80) = 400\text{ m}.

Longueur d'une rangée sans portail = 400 - 4 = 396\text{ m}.

Longueur totale pour 3 rangées = 396 \times 3 = 1\ 188\text{ m} de fil de fer.

Solutions for Section F (Task F-1: Solar Water Tower)

1. Base area: A = \pi r^2 = \frac{22}{7} \times 1.4 \times 1.4 = \frac{22}{7} \times 1.96 = 22 \times 0.28 = 6.16\text{ m}^2. [2 marks]
2. Volume of cylinder: V = A \times h = 6.16 \times 3 = 18.48\text{ m}^3. [2 marks]
3. Capacity in litres: 18.48 \times 1\ 000 = 18\ 480\text{ litres}. [1 mark]
4. Pumping volume in 8 hours:
   • Operating time in minutes: 8 \times 60 = 480\text{ minutes}.
   • Water pumped: 480 \times 25 = 12\ 000\text{ litres}. [2 marks]
5. Comparison and conclusion:
   • Water pumped (12\ 000\text{ L}) exceeds the daily requirement of 4\ 500\text{ L} (12\ 000 > 4\ 500).
   • Excess water generated: 12\ 000 - 4\ 500 = 7\ 500\text{ L}, which can be stored in the 18\ 480\text{ L} tank.
   • Conclusion: Yes, the solar pump supplies more than sufficient water for the crops. [2 marks]

Solutions for Section G-2 (CEP Arithmetic Test)

1. *Recette totale de la vente des bœufs* : 8 \times 275\ 000 = 2\ 200\ 000\text{ CFA F}. [2 pts]
2. *Dépôt d'épargne CLCAM* : 2\ 200\ 000 \div 2 = 1\ 100\ 000\text{ CFA F}. [2 pts]
3. *Reste après épargne* : 1\ 100\ 000\text{ CFA F}.

Dépense pour aliments : \frac{1\ 100\ 000 \times 3}{5} = 660\ 000\text{ CFA F}. [2 pts]

4. *Somme restante disponible* : 1\ 100\ 000 - 660\ 000 = 440\ 000\text{ CFA F}. [2 pts]
5. *Longueur de grillage achetée* : 440\ 000 \div 1\ 760 = 250\text{ mètres}. [2 pts]

End of Workbook 1 (CM2 / MYP 1).